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SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: **Artificial Intelligence** **COURSE CODE: CSA17**

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. *(BL3)*

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

I Rohit Kumar.S (192312214) *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

S. Rohit K.

15/20

B

	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**	**Max Marks**
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.

2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

CSA 1708 . AI

CO1 - AT3

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1) Smart Home Automation using intelligent

Introduction

A Smart home automation system controls appliances such as lights, fans, air conditioners, heaters.

The system receives real time input from sensor like motion detectors, temperature sensor cameras and clocks

Types of Reflex Agent:

1. Simple reflex Agent:

Makes decision only based on the current percept using predefined IF - THEN rules

eg:- if motion detected -> Turn ON lights

if no motion detected -> Turn OFF light

advantages:- • very fast

• Easy to design

• Low computational cost

2. Model Based Agent

Maintains a internal model memory of the environment and keeps track of previous state

eg:- * Bedroom light is alread on.

* Ac is running

Advantages :-

* Makes better decision

* Uses Memory

3. Goal Based Agent

Agent selects action that help acheive a predefined goal.

goals:- * Maintain temperature at 24'c

* Minimize electricity usage

4 Utility ~~Based~~ Agent: agent choose the action ~~that~~ provides the highest overall benefit

Working : Temperature = 27'c

Option 1 :- turn on Ac   Option 2 open window + Fan

Conclusion:- Although each intelligent agent has unique strength it ensures max comfort minimum energy consumption.

2. Robot navigation.

Introduction:

A Robot is placed inside an unknown maze must search for a path from the Start position. to exit.

* Uniform Cost Search
* Iterative Deepening Search

Uniform Cost Search (UCS)

Expands the node with the lowest. cumulative path cost

$g(n)$ = Actual path cost from Start node

Working: Different path have different cost

$S \rightarrow A \rightarrow G$ = cost 8

$S \rightarrow B \rightarrow C \rightarrow G$ = cost 5

UCS selects

$S \rightarrow B \rightarrow C \rightarrow G$ because cost is minimum.

advantage:
* finds optimal path
* Complete Algorithm

Iterative deepening Search (IDS)

This combines the advantage of

DFS & BFS

eg:- Dept 0

S

Depth 1

S
A
B

Depth 2

S
A
B
C
D

(until goal is reached

Advantages.

* Low memory usage
* Complete
* Suitable when goal depth is unknown

Real world Application

* Rescue Robots
* Indoor service robot
* delivery Robots

Conclusion

UCS is ideal for finding the least cost path in weighted environment and IDS is memory efficient for depth